Intelli-Credit AI

AI-Powered Credit Decisioning System

From document upload to CAM generation —
fully automated, explainable, and auditable.

📄 Ingest | 🔍 Research | 🏦 Score | 📋 CAM

The Challenge with Traditional Credit Appraisal

SLOW

5–15 days per appraisal

Committee scheduling delays

Backlogs during peak periods

SUBJECTIVE

Different analysts, different conclusions

No standardized scoring

Promoter research is hit-or-miss

MANUAL

50–200 page reports read manually

Hand-written CAM documents

No audit trail

"Banks spend thousands of analyst-hours each year on appraisals that could be automated."

Intelli-Credit AI: End-to-End Automation

📄 **Upload Docs** → 🔍 **Research Company** → 🏦 **Score & Decide** → 📋 **Download CAM**

- Processes PDF financial documents instantly
- Scours the web for risk signals, litigation, and sector data
- Scores on the internationally-recognised Five Cs framework
- Generates a professional Credit Appraisal Memo (Word + PDF)

⚡ **Full loan appraisal in under a few minutes**

Three-Stage Pipeline Architecture

1 INGESTION

`POST /ingest/document`

- Gemini 2.5 Flash
- PyMuPDF / pandas

→

2 RESEARCH

`POST /research/analyze`

- GPT-5-nano (AIPipe)
- Serper API + trafilatura

→

3 RECOMMENDATION

`POST /recommend/`

- Analyzes research and generate CAM
- python-docx / docx2pdf

AUTO-CHAINED: Stage 1 → 2 → 3 pipeline is triggered automatically on document ingest

`GET /Health` `POST /ingest/document` `POST /research/analyze` `POST /recommend/` `GET /recommend/cam/{f}`

STAGE 1

Smart Financial Document Parsing

What It Does

- Accepts: Annual Reports (PDF), Bank Statements, GST Returns Report
- Keyword scoring filters noise pages (CSR, Chairman's letters, awards)
- Converts raw text → structured JSON automatically with all financial data and research done by research agent
- Auto-triggers Stage 2 (Research) immediately on completion

Structured Output

Revenue	**₹ 45 Cr**
Net Profit	**₹ 6.75 Cr**
Net Worth	**₹ 22 Cr**
DSCR	**1.62**
D/E Ratio	**0.73**
Profit Margin	**15**
Volatility	**Low**
Anomalies	**None detected**

Analyzing with AI

Detecting anomalies & red flags...

- ✓ Uploading document to server...
- ✓ Parsing financial data with AI...
- ✓ Extracting key financial metrics...
- Detecting anomalies & red flags...
- Generating web search queries...
- Searching company intelligence...
- Fetching & analyzing news articles...
- Running LLM risk signal extraction...
- Compiling research report...
- Almost there...

This may take 60–120 seconds. Please wait...

AI-Powered Analysis

Credit Risk Assessment

Upload financial documents or run research-only intelligence gathering — then get an AI-generated credit score, decision, and CAM report.

With Documents | Research Only

COMPANY INFORMATION

Company Name *

e.g. Tata Motors Ltd.

Sector *

Select sector...

Location / City

e.g. Mumbai, Maharashtra

Promoters / Key Personnel

Enter name and press Enter or click Add

+ Add

DOCUMENT UPLOAD

Financial Document *

Document Type *

Annual Report

STAGE 2

6-Stage External Intelligence Pipeline

1 Query Generation

Targeted searches: fraud, litigation, regulatory actions, promoter checks

2 Google Search

Serper API retrieves 20+ unique results per query batch

3 Article Extraction

trafilatura cleans article text; top 3 successful results kept

4 LLM Risk Analysis

GPT-5-nano chain-of-thought analysis per article

5 Signal Consolidation

Deduplicated JSON risk signals with severity scores 1–5

6 Structured Summary

Research summary, news, promoter risks, sector trends

📋 Every stage logged → full pipeline transparency returned in API response for audit

Research Analysis

Mayur Uniquoters · Manufacturing

Continue to Loan Details

RESEARCH SUMMARY

Mayur Uniquoters faces governance risk from management turnover, including the resignation of an SMP and new appointments in HR leadership. It has expanded into a multi-plant synthetic-leather group serving automotive OEMs and export markets, and recently announced a ₹40 crore buyback that modestly affects liquidity. Collectively, these factors imply moderate credit risk from governance changes, customer/concentration exposure to the auto cycle, and liquidity implications from capital allocation.

RISK SIGNALS 3

Medium | reputational | Entity: **Mayur Uniquoters Ltd** | Analysis 1: Change in Management (SMP resignation and leadership appointments)

Resignation of Senior Management Personnel ('SMP') w.e.f. December 06,2025

Medium | industry | Entity: **Mayur Uniquoters Limited** | Analysis 2: Customer concentration and auto/exports exposure

exports into more than 18 countries and the major exports are made to automotive industry.

Low-Med | industry | Entity: **Mayur Uniquoters Limited** | Analysis 3: Buyback

Buyback Amount: ₹ 40 Cr.

COMPANY NEWS

- Expanded from a single-plant PVC coated fabric maker to a multi-plant synthetic-leather group serving automotive OEMs,

PROMOTER RISKS

- Resignation of Senior Management Personnel (SMP) signals governance/turnover risk with potential

SECTOR TRENDS

- Revenue is tied to the automotive sector and global exports, exposing the company to auto-cycle downturns and approvals from

driver.

EXTRACTED FINANCIAL DATA

Revenue	Profit	Net Worth	Debt / Equity
₹834.83 Cr	₹122.47 Cr	₹867.10 Cr	0.02

Current Ratio	Profit Margin
8.76	15.0%

ANOMALIES DETECTED

Medium **Subsidiary Underperformance** · Financial Year Ended March 31, 2024
Mayur Tecfab Private Limited, a wholly-owned subsidiary, reported negative reserves and surplus (-44.62 Lakhs INR) and a net loss after tax (-16.41 Lakhs INR) for the financial year ended March 31, 2024, indicating underperformance.

Pipeline Transparency Logs (click to expand)

query_generation 2026-03-12T16:34:57.623270+00:00


```
{
  "queries": [
    "Mayur Uniquoters India news",
    "Mayur Uniquoters fraud investigation",
    "Mayur Uniquoters litigation India",
    "Mayur Uniquoters regulatory action",
    "Mayur Uniquoters financial trouble",
    "Manufacturing industry outlook India",
    "Manufacturing sector risks India",
    "Rajesh Shah fraud OR investigation OR litigation",
    "Mayur Uniquoters Subsidiary Underperformance Financial Year Ended March 31, 2024"
```


search_results 2026-03-12T16:35:12.099093+00:00


```
{
```

Risk Signal Classification Framework

Risk Type	Examples	Severity
Fraud	ED investigation, fraud FIR filed, Wilful defaulter list	⚠️ **Critical (5)**
Regulatory	RBI penalty, SEBI order, tax authority notice	⚠️ **Critical (5)**
Litigation	Major court case, arbitration, DRT proceedings	🔴 **High (4)**
Industry	Sector collapse, macro headwinds, policy reversal	🟡 **Moderate (3)**
Reputational	Negative press, social media controversy	🟢 **Low (1–2)**

⛔ **AUTOMATIC HARD REJECTION**

Any single Severity-5 signal triggers immediate rejection — regardless of credit score

STAGE 3

Transparent Five Cs Credit Scoring (0 – 100)

max 25 CHARACTER

- Risk signals (severity-weighted)
- Promoter background checks
- No Severity-5 → full points

max 25 CAPACITY

- DSCR ≥ 1.5 → 10 pts
- Profit margin ≥ 15% → 8 pts
- Cashflow volatility → 7 pts

max 20 CAPITAL

- D/E ratio ≤ 1.0 → 10 pts
- Net worth ≥ 3× loan → 10 pts
- Penalised above thresholds

max 15 COLLATERAL

- LTV ≤ 50% → 15 pts
- LTV ≤ 70% → 11 pts
- LTV ≤ 80% → 7 pts | > 80% → 3 pts

max 15 CONDITIONS

- Sector trend baseline: 10 pts
- Positive keywords: +5 pts
- Negative keywords: −5 pts

TOTAL
0 – 100

Credit Decision

Mayur Uniquoters

New Analysis

45
/100

Credit Score
Weak Credit Profile

FIVE CS BREAKDOWN

Character	6/25
Capacity	16/25
Capital	20/20
Collateral	3/15
Conditions	0/15

CONDITIONAL APPROVAL

Primary Reason: Conditional with credit score 45/100. Recommended: ₹4,000,000 at 16.0% p.a. for 24 months. Strongest factor: Capital (20 pts). Key constraint: Conditions (0 pts).

RECOMMENDED AMOUNT	INTEREST RATE	TENOR
₹40.00 L	**16% p.a.**	**24 months**

Score Explanations (Full AI Reasoning)

Credit Assessment Memorandum (CAM)

AI-generated comprehensive credit report in Word and PDF format

Download Word (.docx)

Credit Decision Logic — Score Bands

Score	Status	Loan Amount	Rate	Tenor
75 – 100	✅ APPROVED	100% requested	10% p.a.	60 months
60 – 74	✅ APPROVED	80% requested	12% p.a.	48 months
50 – 59	✅ APPROVED	60% requested	14% p.a.	36 months
40 – 49	⚠️ CONDITIONAL	40% requested	16% p.a.	24 months
< 40	⛔ REJECTED	—	—	—
Any Sev-5	⛔ HARD REJECT	—	—	—

Worked Example — Rajesh Industries Pvt Ltd

📥 Application Input

Company:	**Rajesh Industries Pvt Ltd**
Sector:	**Manufacturing**
Promoter:	**Rajesh Kumar**
Document:	**Annual Report 2025 (PDF, 120 pp)**
Loan Ask:	**₹ 50,00,000**
Collateral:	**₹ 80,00,000 (Residential property)**

📊 Key Financials Extracted

Revenue	**₹ 45 Cr**
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Net Worth	**₹ 22 Cr**
DSCR	**1.62 ✓**
D/E Ratio	**0.73 ✓**
Profit Margin	**15 % ✓**
Volatility	**Low ✓**
Anomalies	**None detected**

Worked Example — Five Cs Scoring Result

CHARACTER	**21 / 25**	84%
↳ Minor tax dispute — −4 pts		
CAPACITY	**25 / 25**	100%
↳ DSCR 1.62 + 15% margin + Low volatility		
CAPITAL	**20 / 20**	100%
↳ D/E 0.73, Net worth 4.4× loan		
COLLATERAL	**11 / 15**	73%
↳ LTV = 62.5% → adequate cover		
CONDITIONS	**13 / 15**	87%
↳ Manufacturing PMI at 14-month high		

TOTAL SCORE **90 / 100 (Band: 75–100)**

Worked Example — Decision & Generated Output

✅ APPROVED

Loan Amount	**₹ 50,00,000**
Interest Rate	**10.0% per annum**
Tenor	**60 months (5 years)**
Score	**90 / 100 (Band: 75–100)**

Strong capacity, clean promoter profile, positive sector outlook

📁 Output Files Generated

📄 `Rajesh_Industries_a1b2c3_CAM.docx`

↳ 7-section Word document

📋 `Rajesh_Industries_a1b2c3_CAM.pdf`

↳ PDF export — ready to share

Download endpoints:

```
GET /recommend/cam/{filename}
GET /recommend/pdf/{filename}
```

Generated Credit Appraisal Memo (CAM)

1 Executive Summary

AI-generated credit proposal overview

2 Applicant Profile

Company, sector, promoters, loan request

3 Five Cs Analysis

Each C: narrative + score table

4 Financial Analysis

Key metrics: revenue, ratios, DSCR

5 Risk Matrix

All risk signals with evidence + severity

6 Credit Score Summary

Tabulated Five Cs scorecard

7 Recommendation

Final decision with full reasoning

Formatted: python-docx | Exported: docx2pdf

Built On Modern, Production-Grade Stack

🤖 AI / LLMs

- Google Gemini 2.5 Flash
- GPT-5-nano
- OpenAI-compatible API
- Pydantic v2 validation

🌐 API & Web

- FastAPI + Uvicorn
- Serper API (Google Search)
- trafilatura (scraping)
- httpx (async HTTP)

📄 Document Processing

- PyMuPDF (fitz)
- python-dotenv (config)
- Pydantic v2 schemas

📁 Storage & Output

- Local filesystem storage
- python-docx (Word gen)
- docx2pdf (PDF export)
- storage/cam/ directory

Explainability at Every Step

Transparent Scoring

- Every point awarded or deducted has a written reason
- Rules are fixed and inspectable in code
- No black-box AI scoring
- Weakest C identified and named

Research Logs

- Every stage logged: queries, results, extracted text, analysis
- Pipeline logs returned in API response
- Article sources cited with evidence quotes
- Timing per stage included

Documented Decisions

- CAM documents decision with full narrative reasoning
- Score explanation per parameter
- Severity explanation for each risk signal
- Complete audit paper trail

"Explainable AI is not optional in regulated banking — it is mandatory."

Security by Design

API Key Isolation

Stored in .env only — never in code or version control

Safe File Handling

All uploads renamed to UUID; no user-controlled paths reach filesystem

Input Validation

Pydantic v2 validates and sanitises all API inputs at every boundary

Request Timeouts

300-second timeout on all external HTTP calls; no hanging requests

Storage Isolation

Uploads and generated CAM files kept in separate directories

No SQL / No Injection

No database used; SQL injection attack surface eliminated entirely

Immutable Audit Trail

Every research decision and score reason persisted in CAM document

Getting Started in 3 Steps

1 Clone & Install

```
git clone
https://github.com/SujalPradhan/Intelli-Credit-AI

cd Intelli-Credit-AI

uv venv

uv pip install -r
requirements.txt
```

2 Configure .env

```
GEMINI_API_KEY=your_gemini_key
SERPER_API_KEY=your_serper_key
AIPIPE_API_KEY=your_aipipe_key
```

3 Run

```
python
frontend_server.py
# It serves frontend on
http://localhost:8000/ui
# Swagger UI is also
available immediately
on /docs
```

System Requirements:

- Python 3.11+
- Windows OS (for docx2pdf)
- uv (recommended) or pip
- API keys (Gemini · Serper)

What Intelli-Credit AI Delivers — & What's Next

✅ Delivered Today

- Full PDF document parsing
- 6-stage external research pipeline
- Deterministic rule-based Five Cs scoring (0–100)
- Band-based decision logic with hard-rejection rules
- Professional CAM generation (.docx + .pdf)
- Interactive UI @ http://localhost:8000/ui
- Swagger UI for interactive testing

⬜ Potential Roadmap

- Web dashboard / front-end UI
- Multi-language document support
- Database persistence (PostgreSQL)
- Role-based access (maker-checker)
- Bureau data integration (CIBIL / CRIF)
- Portfolio-level risk dashboards
- Cloud deployment (AWS / Azure)